

Section 2

Digital Transformation across Industries

Chapter 2

Digital Banking 2.0

Banks have been attacked by digital natives. However, they still command high market share, driven by customer stickiness.

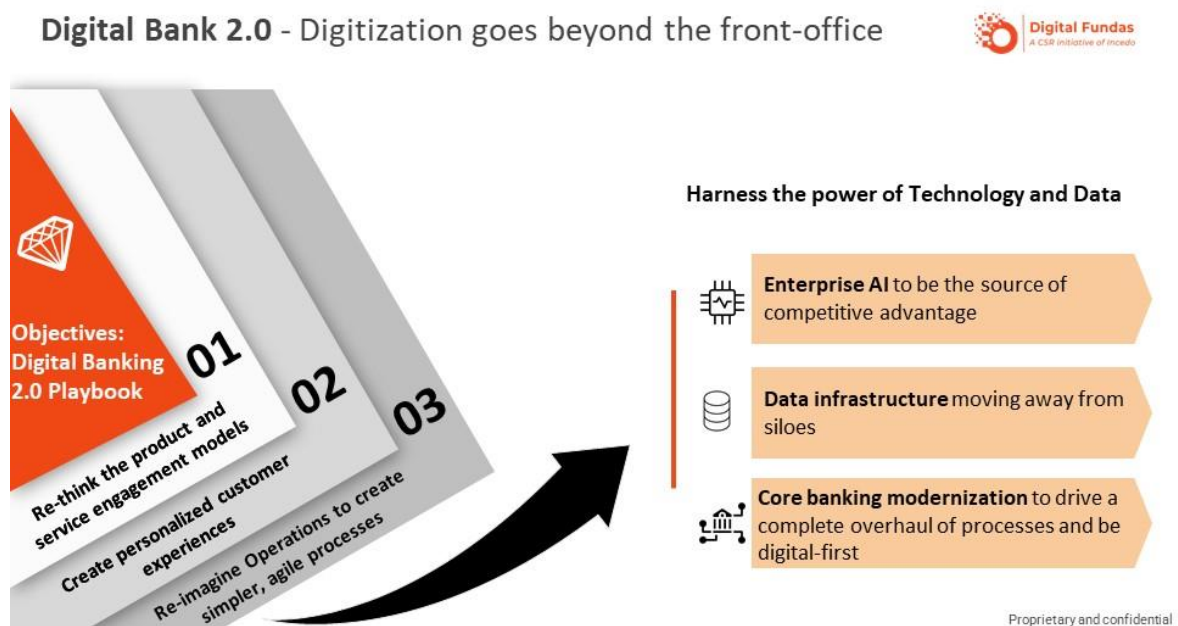
Most of the banks' efforts have been centered on digitizing the front-end and developing a better system of engagement.

Whereas the core issues of their legacy system of records and business processes – which are stuck in a 'control' mindset – still persist.

Key Takeaways

Digital banking focus areas - product and services engagement, personalized customer experience, operations, technology, infrastructure modernization, Enterprise AI, Data

Lesson 1: Digital Transformation 2.0: Key building blocks



Traditionally, banks have focused on building trust and security with their distribution network of branches. They have, by and large, not been able to keep

pace with the improvements in customer experience observed across other consumer businesses. As Nubank has shown in Brazil, banks need to fundamentally rethink what it will take to pivot to a more customer-centric business model, aggressively go beyond their branch-based distribution networks, and at least invest in operations transformation which can bring back the returns to scale. This will require a tremendous shift in focus from strategy to execution across all dimensions – of people, processes and technology.

I believe the Digital Banking 2.0 playbook must include the following objectives:

1. **Rethinking of the product and service engagement model** to move away from product-centric silos to a more holistic customer-centric approach.
2. **Creation of personalized customer experiences** by servicing customers in the channel of their choice in a truly multichannel world.
3. **Re-imagining of operation** to create simpler, agile processes that are able to adapt to rapidly evolving customer engagement needs

Data and technology will undoubtedly be the backbone on which these objectives will hinge while the execution process is on.

A comprehensive technology strategy for a bank must include the following:

- Enterprise AI – to design, train and deploy the next generation of analytical models.
- Data infrastructure – to power the AI models.
- Core banking modernization – to not just enable process automation but also to drive actions based on AI model recommendations

Lesson 2: Product and services engagement

In most banks, products and services are designed and executed along functional product lines. While this drives a product-centric efficiency mindset, it misses out on a crucial factor: customers do not always think of their financial journeys as falling into such well-defined brackets. For instance, a family might have to make a choice between home improvement and college tuition, both of which are driven by a combination of factors related to the customer's life journey. In this scenario, in all likelihood, two different functions within the bank would individually design offers and target the family, not factoring in the trade-offs that the family needs to make in choosing either option. This comes with another paradox. The biggest competitors for banks today, the fintechs, usually offer specific products. The bank would, in all likelihood, be competing with two different fintechs – one in home equity and another in education loans.

Today, banks can convert this challenge into an opportunity: what if the bank takes a complete life cycle-journey view of the customer and uses that to define bundled product offerings that can combine the home improvement and education loans? A data-driven approach would look something like this:

- Leverage AI-based choice modelling to predict customer responses to different offerings, standalone and bundled.
- Use scenario modelling to quantify the value and cost trade-offs both for the bank and the customer base.
- Tailor offerings specific to customer micro-segments to drive maximum economic value.

These data-driven recommendations could then serve as the basis for customer relationship managers to have a truly deep and meaningful engagement with their customers. This cross-product, customer-centric approach has the potential to create a competitive advantage for banks over the product-specific fintechs.

A leading Cleveland-based bank introduced three new banking products: no-fee overdraft protection, a 2 percent cash-back credit card, and a debit-linked savings programme, all bundled together as a solution to reward customers for working towards their financial goals. Danske Bank, a northern European bank, has gone a step further and offers a housing app to its customers to help them navigate the house-buying process, including the loan applications, approval and disbursement process. It goes even further, offering customers home equity loans to finance the purchase of appliances, etc. In both these examples, the banks decided to step out of their traditional product mindset to look at the overall customer journey and design customer-centric solutions.

Lesson 3: Personalized customer experiences

Two of the biggest demographic trends in consumer banking in recent times have been:

- **Slowing down of new customer acquisitions:** In mature economies like the US, the deposit base has levelled off over the last several years. This, in turn, has meant that banks have to rely on their existing customer base to deliver growth.
- **Channel preference by age group:** While the baby boomers have not wholeheartedly adopted digital banking (although that is changing in the post-pandemic world), millennials and the subsequent generations, by and large, prefer to almost exclusively bank digitally.

Both these secular trends have meant that a key growth driver for banks is going to be increasing revenue per customer – which is why cross-selling and up-selling are

becoming key elements of customer engagement. As more and more customer interactions shift to digital channels, there is an opportunity for traditional firms to learn from tech firms. For instance, Amazon attributes 35 percent of its sales to cross-selling⁷ – it has achieved this by creating hyper-personalized customer experiences on the back of data and AI. Taking a leaf from their playbook, here is how a bank might want to approach the personalization challenge:

- FAANGs pioneered the idea of ‘**Segment of 1**’ – intending to move away from the traditional idea of broad segments, which limited personalization – to develop micro-segments with a two-pronged strategy. This strategy enabled them to engage with customers with a high degree of precision, which in turn has significantly expanded the customer relationship and share of wallet for them:
 - Identify transactional and behavioural attributes that help define customer characteristics to a high degree of personalization. For a bank, this would be along the multiple dimensions of wealth and income, financial acumen and digital sophistication.
 - Approach segmentation as dynamic. Over a customer’s life journey, the need states and capabilities (especially financial status), and hence, behaviours, continue to evolve all the time. And with that, the segments will need to be refined with ever-increasing frequency.
- As banks’ understanding of customers deepens with micro- segmentation, it becomes possible for them to design personalized offers tailored to specific customer micro-segments and layered with customer-need states. For a bank facing limited net growth in new customers, it is critical to design and offer products and services which cut across product lines. Personalized engagement needs to be addressed in the following dimensions:
 - What is the most relevant product/service?
 - What is the right channel? A bank would need to factor in the possibility that customers can switch channels within specific interactions. For example, a customer might respond to a digital offer by calling into the call centre and/or walking into a branch. The bank needs to have the capability to provide a seamless experience to the customer through these channel hops.
 - What is the right engagement frequency? Unlike in retail shopping, customer interactions with banks are relatively low-frequency, with a high-value transaction happening only once in a while. This requires the bank to be able to drive the right frequency of engagement across its channels to be able to truly capture the customer’s attention at the right time with the right product/service.
- And last but not least, the traditional notion of customer lifetime value (CLV) was pioneered in the retail industry and embraced by banks over the last two decades as an underpinning for loyalty programmes. As loyalty, especially among gen-Zs, wanes, it is time for banks to start moving away

from attempting to predict the total lifetime value of customers and instead focus on the ‘next best action’. The goal is to predict immediate actions that are likely to have the highest customer impact and hence maximum response rates.

- This is becoming even more important in the highly volatile and uncertain economic environment that we have today. For instance, in the current sluggish economy, a bank should focus on protecting its credit customers by driving actions sensitive to their need states. A proactive adjustment offer of loan refinancing with the focus on the customer’s current situation, as opposed to a reflexive reaction to an eroding CLV, for example, could go a long way in building customer loyalty.

Lesson 4: Operations

Digital banks have exposed two main operational gaps in traditional banks:

- From back-end centralized processes (e.g., loan processing) to distributed processes, (e.g., account opening at a branch) everything continues to be largely manual. And whatever automation does exist is typically at the task level; the processes themselves haven’t changed over all these years. Digital banks, meanwhile, have shown that vastly simplified processes are possible.
- The scale advantage that larger retail banks enjoyed over the years, with their shared overheads for IT, infrastructure, etc., may not exist for too long. Digital banks have shown that productivity improvements led by technology (not just RPA, but cognitive AI too) are possible across a bank’s processes.

Together, these two factors are not just impacting the cost structure of banks, but also creating vastly superior customer experiences. I see three big areas of opportunity for transforming banking operations:

1. **Self-service and straight-through processing:** Processes were defined with a ‘risk and control’ mindset that focused on minimizing operational risk at the cost of speed and flexibility. As the digital natives from banking to retail have demonstrated, it is possible to reimagine the processes with customer experience as the primary goal, and then define technology-enabled risk guard-rails for exception processing. This is possible, especially because customers increasingly prefer digital self-service options. One success story here is that of cheque deposits: this process has moved from being a manual, in-branch activity to a near- flawless straight-through processing experience. Many more such opportunities do exist – it will take a rigorous approach to identify them, define the SLAs that could be impacted, and drive the solutions to enable them.
2. **Intelligent automation:** Task automation through RPA (e.g., bots that auto-populate forms) has, by and large, run its course and delivered point

efficiency gains. The next wave of automation is a combination of AI-enabled intelligent task automation (e.g., document scanning to extract data for a variety of document verification activities) and workflow-enabled end-to-end process automation (e.g., customer onboarding, verification and disbursement of consumer loans). These not only create great customer experiences by reducing the time and effort required but can also free up time for relationship managers to engage with their customers in deeper and more meaningful ways.

3. **Redefinition of operating processes:** This would be to take a more dynamic and flexible approach in response to a rapidly changing macro-environment and its impact on the customer base, as evidenced by the events of 2020. A leading consumer bank is relooking its consumer portfolio-level risk assessment in light of the economic chaos triggered by the Covid-19 pandemic. Instead of relying on payment history alone to assess risk of default, the bank is looking to glean underlying behavioural factors from data points like contact history – not just at the number of contacts and contact channels, but also the contact sequences – to assess the default risk scores on a daily basis. This not only requires building AI models that can supplement the risk-scoring models but also means suggesting interventions for the field agents.

Nitin's conclusion:

I believe the Digital Transformation 2.0 journey has just begun. Over the next few years, I expect to see a dramatic change in digital retail banking. In much the same way that computing technology with algorithmic trading and automation helped drive radical change in banking capital markets, technology, I believe, will be a key enabler in driving transformation in consumer banking too.

Lesson 5: Technology

Banks have traditionally been big technology spenders. Some of the biggest banks have spent billions of dollars on digital transformation initiatives. The results have been less than stellar. In a recent survey, 75 per cent of investors who were asked, 'How confident are you that banks' digital transformation strategies will be effective?' replied with either 'skeptical' or 'too soon to tell'. On the other hand, these very banks have, on average, spent close to 5 per cent of their revenues on transformation programmes, which amounts to hundreds of millions of dollars.⁸ It feels like a digital technology trap – lots of investments with little to show by way of outcomes. And, as expected, technology spending varies across banks. While the biggest banks spend around 0.4 per cent of their asset base on technology (that alone is a whopping \$11.4bn for JP Morgan Chase), the mid-tier banks (with assets of between \$500m and \$5b) clock in at 0.22 per cent, or about half the proportion

of asset base the big banks put in. Generally speaking, while the big banks are struggling to show results from their tech investments, the smaller banks are, in all likelihood, not spending quite enough.

Suffice it to say that big or small, banks need to think hard and drive the right set of technology priorities to stay ahead in their digital transformation journey. This they must do by pushing on the key building blocks outlined above. And, in my experience, any digital technology strategy must include meaningful investments in three areas: infrastructure modernization (table stakes), enterprise AI, and data.

Lesson 6: Infrastructure modernization

The banking industry was among the first to drive large-scale investments in IT infrastructure. And, as it often happens, being the first mover is both an asset and a liability. While it has served to drive productivity improvements over the years for the banks, aging technology and rigid infrastructure have become more of a liability for them in recent years. This has spurred the banks to invest in complete technology infrastructure upgrades. While this is certainly necessary, it is not sufficient to drive competitive advantage, especially when the competition consists of fintechs and digital banks.

Infrastructure modernization is underway and is likely to be a big investment area for banks for several years to come.

Some of the key factors that will determine the long-term success of their core infrastructure modernization are:

- **Cloud first:** Most CIOs have accepted the fact that managing IT infrastructure is not a bank's core competency and that cloud migration and enablement is the most important imperative for banks to be able to modernize their technology infrastructure.
- **Cloud security:** The rush to cloud migration needs to be balanced with the right cloud security, especially given the sensitivity of data banks have to handle (banks continue to remain the favourite target of malicious hackers). Most banks are opting, rightfully so, for a hybrid cloud infrastructure.
- **Modernization is a journey:** Banks are saddled with multiple systems, starting with the core banking platform and layers of other applications. This has created a patchwork of data sharing and integration built over years. Starting with core banking, upgrading all of the systems to a modern cloud infrastructure is a multi-year journey. Through these phased, multi-year migration programmes, it would be necessary to go through phases of data duplication across the legacy (on-premise) and modern (on- cloud) systems. This forced inefficiency needs to be managed well with integration governance.

Interoperability: One of the key – if not the most important – elements of a modern, distributed application ecosystem is the definition and adoption of architecture and interoperability standards built on modular architecture elements and well- established protocols for data sharing through APIs.

Lesson 7: Enterprise AI

Organizations have run AI projects for several years now. But they are mostly isolated examples in the areas of marketing and risk management. These projects were typically limited by data and compute constraints and are nowhere near to the large, sophisticated AI engines that power the fintechs and bigtechs today. Banks can no longer afford to remain at the level of piecemeal projects, at least if they hope to stay relevant and compete against the digital natives in the coming years.

It is essential that banks today invest in enterprise AI with the following clear capabilities to support a wide range of use cases across products and services, customer engagement and operations:

1. An enterprise AI platform that can support some of the latest in AI techniques, all the way to deep learning. More importantly, the platform should have the ability to incorporate the latest advances that are coming out of academia, open-sourced from FAANGs, etc.
2. AI is compute-resource hungry, and nothing can be more frustrating than a constraint on resources. To deliver on scale and performance, it is inevitable that the enterprise AI platform should be cloud based.
3. It is always worth remembering that AI is not for AI's sake – the goal is to deliver business outcomes. And for that to happen, there needs to be a clear way of taking AI recommendations and using them to drive business processes. This requires the enterprise AI platform to:
 - Operationalize the AI models – starting with the data pipelines, training, deployment of models, and ongoing monitoring of the models.
 - Integrate the model outputs with operational systems to integrate seamlessly into the overall decision-making process.
4. There needs to be recognition of the fact that AI has its roots in the open-source industry, from shared datasets to shared algorithms. A robust AI ecosystem in the enterprise must be able to replicate this democratization across functional silos. That is critical to enable a culture of reuse and sharing of best practices; and perhaps, more altruistically, to build and foster a learning ecosystem with the enterprise data science community.
5. And last but not least, the need for strong AI governance cannot be understated. One of the biggest challenges of AI relates to its ethics. Enterprise AI needs to take ownership of the following ethical dimensions:

- Data privacy and security: Access to data must be balanced with the enterprise security guidelines, especially when it comes to customer-sensitive data.
- Algorithm bias: Consumer banking is especially sensitive to this – particularly when it has material implications for specific population segments. There needs to be constant monitoring of algorithms to guard against bias.

Lesson 8: Data

Data may or not be the oil of the twenty-first century, but it certainly is the fuel that powers AI. And yet, organizations have failed to devote the necessary attention and resources to build data as a strategic asset. And if there is a single reason why AI has not delivered the promised value, it is this myopic view of data. For the longest time, organizations have treated data as an afterthought: the recent excitement about data lakes and the investments in them haven't yielded much either.

The situation with banks is no different. If there is one (more) lesson to take from the tech firms, it is that for data to truly generate economic value for the firm, it has to be treated as front and centre of the business and operating model. As banks, big and small, are finding out, this is easier said than done. The biggest constraint is legacy. The current data infrastructure and data models have evolved through functional silos, archaic systems and decades of accumulation of disparate systems through mergers and acquisitions. To add to that, organizational data fiefdoms have resulted in patchy (at best) progress in standing up a data ecosystem that can support AI to drive digital transformation.

This is why I believe that data must be one of the top organizational priorities for banks if they expect to survive and thrive in the fintech onslaught. Some of the key data priorities that banks need to embrace right now are:

1. **Cloud data platform:** The last decade saw organizations splurge millions on Hadoop systems, swept by the 'big data' narrative. That is unravelling rapidly, as the on-premise Hadoop infrastructure is unable to scale, both in performance and costs. It is time to move away to the cloud, and not just from the cost and scalability points of view. The cloud also offers enterprises the ability to innovate, especially with event logs and streaming data, which are the backbone driving real-time digital customer engagement.
2. **Data pipelines:** Any data infrastructure is only as good as the data. It may sound obvious, but the reality is that most banks have not done enough at all to move data from multiple legacy systems to a centralized source. Banks have (and continue) to burn millions of dollars on building enterprise data lakes with the promise of creating a curated, centralized data repository for all analytical needs. Clearly, this has not happened with any degree of success (not just in banks, but across industries) – largely because the time,

effort and focus required are just not available in most organizations. Eventually, the result is less of a clean data lake and more of a muddy data swamp. In any case, it is important to invest in building and maintaining the flow of data into the cloud-based centralized repository for it to be even useful.

3. **Data harmonization:** This involves more than just transfer of data; it needs a complete integration and harmonization roadmap across disparate systems. In a typical bank, customer data is sourced from multiple systems – many organic (e.g. account transactions; loan origination; customer care) and many inorganic (systems acquired from bank mergers and acquisitions). Integration across these systems to create a complete data view is necessary. This is perhaps the hardest challenge to surmount, less from a technical perspective but more from an organizational process perspective.
4. **Data quality:** The GIGO (garbage in garbage out) principle has never been truer than in the digital world. Much like data harmonization, this is a tough problem, and yet a necessary one to solve. In a not-to-distant future, when the banks will need to drive personalized engagement with customers and automate operations internally, good-quality data will be key to ensuring that these processes drive the expected value.
5. **Data for AI:** AI feeds on data, and good AI is one that learns and keeps getting smarter over time. To make this happen, well-documented and reliable training and validation datasets need to be created. They must not just be comprehensive but also have guards against reinforcing algorithmic bias.
6. **Data governance:** None of the above is remotely possible without strong data governance that operates at the bank-level (not functional group level) and takes responsibility for:
 - a. Breaking down the trust barriers to share data across groups
 - b. Establishing a data ethics framework to protect against bias
 - c. Advancing the cause of a centralized, shared data ecosystem with the right security and regulatory compliance

It is obvious but worth repeating that these technology investments in infrastructure modernization, enterprise AI and data are not to be treated as sequential. The real competitive advantage for banks is in the how: the ability to execute on all three fronts, even as they are continuously evolving with and learning from the market dynamics and other macroeconomic factors.

Nitin's conclusion:

Digital transformation in retail banking is at an inflection point, and we can also expect this to accelerate significantly in the coming years, even as the industry is going through profound changes triggered by the economic uncertainty arising

from the coronavirus pandemic. Success in this difficult environment will require clarity in strategic direction, the ability to make the right bets in technology and people, and, above all, speed and agility in execution. One thing, though, is increasingly certain: the banks of tomorrow will look a lot less like the branch-based networks we have been used to and ever more digital, with a limited physical footprint, offering customers a truly seamless multi-channel experience at every stage of their financial journey.

Section 2

Digital Transformation across Industries

Chapter 4

How Digital is Disrupting Payments

- Payments has been at the forefront of digital transformation, and already FAANGs (Facebook, Amazon, Apple, Netflix and Alphabet) and the Chinese Big Three (Alibaba, Baidu and Tencent) have attacked this space.
- One can expect some big moves in this space as it will see disruption happening at a rapid pace.
- The enterprise payments space will see a change similar to what has already been observed in the consumer-to-consumer segment.

Key Takeaways

Trends that will impact the Payments segment

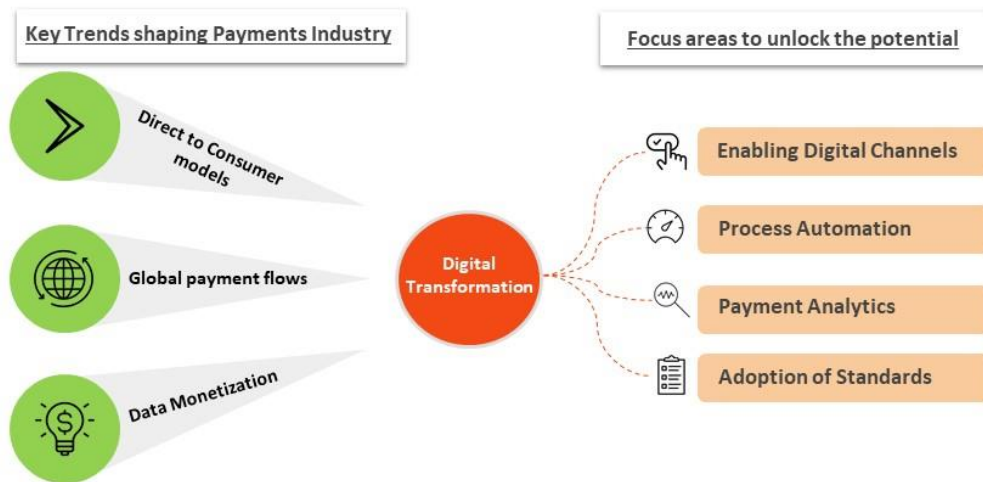
Digital Transformation in Payments: 4 Areas of focus

Lesson 1: Three big trends that are likely to shake up the Payments segment

Nitin's introduction:

To say the payments industry is going through disruption is certainly not a hyperbole these days. The fundamental shifts in how commerce gets done have begun to impact the way payments have been done all these years. On the one side, the payments industry has seen the entry of diverse fintech players, including giants like Facebook and Tencent, in addition to the start-ups that are presenting increased competition for banks and corporations. On the other hand, the threat from fintechs is being further fuelled by rapidly evolving customer expectations, which continue to push the boundaries for the industry as a whole. It is increasingly apparent that the payments marketplace will look fundamentally different a decade from now. There will be new form factors, real-time infrastructure, greater levels of integration with social media and e-commerce, to name a few of the changes. In effect, the revolution that has completely disrupted the consumer payments industry over the last decade or so is finally coming to take into its fold the corporate payments industry, too.

Payments - Disrupt or get disrupted!



Proprietary and confidential

- **Direct-to-consumer models:** As firms across industries move to direct engagement with their customers, it is becoming increasingly necessary to deliver them the same level of digital experiences as the consumer payments industry does.
- **Global payment flows:** Cross-border payments now make up over 10 per cent of all corporate payments, and they are growing.¹¹ These flows are almost always digital in nature, with the added complexity of regulatory compliance and risk management.
- **Data monetization:** Bank treasury services have had the advantage of managing and servicing fund flows between their corporate clients. And as these fund flows become increasingly digital, they have enabled banks to build a data goldmine. Banks are now actively looking to leverage this data to deepen their service offerings.

Banks and corporations have started responding to the call of digital, and the payments processing industry is currently going through a wave of infrastructure modernization. I see significant technology investments by CIOs across firms that are setting the stage for the next wave of digital transformation. The payments industry will look fundamentally different a few years from now. By adopting digital channels, embracing automation, adopting open standards and making smart bets in technology, banks and corporations can emerge as winners in the payments marketplace.

Lesson 2: Making digital transformation happen

As digital transformation initiatives in payments pick up steam, there are four main areas of focus, each of which is important to ensure not just a solid foundation for a digital payments ecosystem, but also to ensure the groundwork for unlocking the revenue potential from treasury and payment operations. This is something yet to be tapped in most organizations:

- **Enabling of digital channels:** Buoyed by the consumer payments industry, there is a rapidly growing array of digital payment channels that need to be integrated into the digital payments service offerings.
- **Process automation:** Payment processes typically span across entities (bank-corporation-consumer), and integration across disparate systems will make for a critical foundation to enable scalable implementations.
- **Payment analytics:** Payment processes have always been data rich, and even more so when digital channels continue to grow. Effective use of the data to make better decisions (e.g., manage risk, prevent fraud) and, furthermore, explore data monetization opportunities, are becoming important.
- **Adoption of standards:** For any multi-entity ecosystem with entities across the globe to scale with technology, it is essential to establish standards. Adoption of open banking standards is essential for digital payments to succeed – and we are at an inflection point, given the increasing adoption of these standards.

Lesson 3: Digital channels

The ubiquitous cheque has been the staple in corporate payments for years now. Despite being the most expensive payment instrument, the cheque has dominated the corporate payment world for decades. It is not just the processing cost of the paper cheque that makes it a burden for banks. It is also a security headache. It is well known that paper cheques are the largest vehicle of payment fraud. All that is changing. And, as it usually happens, this started with the consumer payments business. As the direct-to-consumer models continue to evolve, the B2C payments business is growing rapidly (annual growth rate of 15 per cent led by digital e-commerce). And the digital payment technologies that are coming out of the consumer payments industry (Zelle, Paypal, digital wallets, et al) offer a rich choice for banks and corporates to offer digital experiences to their customers:

- **Disbursement of funds:** Transfer volumes of Medicare/ Medicaid funds by healthcare providers to their members continue to rise and should, by and large, be digital.
- **Refund management:** In a direct-to-consumer world, corporations need to manage refunds to customers from excess payments and product returns. Customers used to instant digital payments from the e-commerce world are expecting a similar experience everywhere.
- **Loyalty/reward disbursements:** As corporations build deep relationships with their customers, they continue to adopt customer engagement strategies from e-commerce retailers. These include cash-back payments and encashing of loyalty points, which need to be executed through digital channels.

A similar revolution is around the corner in B2B payments, with the expansion of consumer-like payment rails, such as digital wallets, in addition to the existing ones like ACH, Wire, virtual cards, etc. We believe the convenience of digital payments is only a starting point. There is so much more to it by way of benefits:

- *Streamlining of payment processes* has a direct impact on working capital management. Trade finance is key to enabling global supply chains, and fintechs are coming up with specific solutions.
- *Cross-border payments to suppliers and subsidiaries* need to stand up to heavy regulatory requirements in addition to managing the risk of fraud.

Digital payments are increasingly the safest alternative. In addition to enforcing compliance, digital channels can ensure transparency of global fund flows.

- Of late, *acquiring a deeper understanding of the supplier ecosystem* has become an important factor (see the section on payment analytics below).

Lesson 4: Automation and Data & Analytics

Automation

70 per cent of corporate treasury and payments professionals list manual and inefficient processes among their top challenges. In addition to their high costs, manual processes are also error-prone, difficult to scale in response to variable volumes, and increasingly susceptible to fraud.

Process simplification and automation opportunities extend across the value chain – from establishing the payment exchange with suppliers (B2B) and consumers (B2C) to creating a variety of services around three-way (PO, invoice and receipt) matching, and all the way to the disbursement of funds through different digital payment channels. Several fintechs see this as a big area of opportunity and are building to this end auxiliary platforms that can integrate with corporate systems and automate the end-to-end processes.

Bank treasury services offer a slew of products and services – from cheque processing to ACH/Wire – to their corporate clients. For instance, a large bank helps one of the largest healthcare providers in the US process over 4 million transactions on an annual basis, covering their entire value chain, from providers – corporate hospitals (B2B) and individual doctors (B2C) – to pharmaceuticals (B2B).

- There is a clear opportunity to digitally onboard these entities onto the payments network in a rapid, secure manner using DIY portals as well as create an 'omni-channel' like experience that minimizes onboarding friction.
- Automating a payment network of this size and complexity is undoubtedly an integration challenge, given the multiple legacy systems at enterprises and, increasingly, ERP systems. With the increasing adoption of API (application programming interface)-based data exchanges, end-to-end automation with multiple payment rails is a necessary building block for digital payments.

Data and analytics

Payment processing through digital channels is data rich: strategies and execution led by analytics on the transaction data can help in a variety of ways: improving revenues, cutting operating costs, detect fraud and other anomalous behaviour.

Risk and fraud analytics: As payments migrate to digital platforms, it is almost inevitable that fraud becomes more sophisticated too. And, as the volume and complexity of payments grow, fraud is becoming just as hard to track, identify and prevent. Fraud prevention will have to move beyond transaction-centric assessment

to leveraging AI for detection and prevention of emerging fraud. Broadly speaking, organizations need to think of payments fraud at two levels:

- **Account fraud:** Digital identity theft is a leading cause of fraud. Methods like ATO (account takeover) and synthetic identity creation can be used to gain access into accounts and siphon funds. Tracking and preventing this requires going beyond the traditional knowledge-based authentication methods to monitor authentication journeys, looking for anomalous patterns.
- **Phantom payments:** Businesses lose significant amounts to fraudulent payments, triggered both by employees (e.g., initiating a phantom payment) or payees (e.g., creating double invoices). Monitoring and flagging them requires a range of methods, starting from rule-based systems (e.g. proximity of transaction requests) to more sophisticated machine learning methods (e.g., payee behaviour risk-scoring and setting of guard-rails sensitive to each risk segment).

Data monetization: A bank managing the payment flows between its corporate clients and their network of suppliers and customers has the unique ability to understand the financial behaviour of all the firms in this network. This can be a powerful tool for the bank to develop targeted strategies to drive superior experience and value for its clients:

- **Working capital optimization:** Banks can help their clients optimize their working capital by forecasting fund flows and using that to transfer the optimal funds into their payment accounts.
- **Service bundling:** Using a combination of behavioural and transactional patterns, banks can help define optimal service bundles for their clients. For example, corporate payments that span multiple countries can be optimized with a combination of exchange-rate hedging and currency-float solutions.

Lesson 5: Adoption of standards

Open Banking

Starting in 2015, when the European Parliament adopted open banking standards (PSD2), there has been a growing momentum in adoption of standards. And, as it happens with standards, this can catalyse innovation and efficiency across the world of payments once they reach a critical mass of adoption. Open banking regulations require banks to open up their systems and data to third-party providers through secure channels. This has the potential to accelerate:

- Seamless transfer of funds between banks, using standards as opposed to relying on the current custom of point-to-point software integrations.
- The capability of corporations with multiple bank accounts across currencies to efficiently aggregate bank account data into a single accounting portal for automated reconciliation, mitigating issues in one of the most complex of payment transactions – cross- border payments.

Blockchain technology

Several financial services firms are increasingly looking to blockchain technology to mitigate the risk of fraud. The three fundamental underpinnings of the technology are distributed ledger, and immutable and permissioned access. Taken together to underpin a payment processing service, they make it possible to trace the entire sequence of wire transfers. Visa launched its B2B Connect Platform based on a private blockchain with the aim of enabling faster cross-border payments. Similarly, a host of banks, including HSBC, BNP Paribas and ING, launched Contour, a blockchain- inspired platform designed to make the \$18-trillion trade finance market more efficient and secure. I expect this space to see a lot more action in the coming years.

After decades of plodding along with archaic systems, the \$2 trillion behemoth that is the global payments industry, is waking up, shaken up by the fintechs (a revolution of sorts that PayPal ignited). And, as it often happens, innovation in one sector rapidly spills over to adjacent areas; the dramatic change that started in consumer payments created the technology building blocks for digital disruption in corporate payments too. Combined with the adoption of standards and, most notably, the maturity of blockchain technologies, the corporate payments industry is primed for a burst of innovation.